

12 Transformers

12.1 self attention processes N inputs of length D to produce N outputs

	weights	biases	
queries	$D \cdot D$	D	so total weights: $3D^2$
keys	$D \cdot D$	D	total bias: $3D$
value	$D \cdot D$	D	

and $N \cdot N$ attention scores.

In a FC shallow network, there would be

$$|weights|: DN \cdot DN$$

$$|biases|: DN$$

12.2 so we can stack them!

$$12.3 \quad \text{or } S_a[XP] = S_a[X]P$$

$$\begin{aligned} S_a[XP] &= U[XP] \cdot \text{Softmax}[U[XP]^T Q[XP]] \\ &= \underbrace{P_V^T + \Omega_V X P}_{= P_V^T P + \Omega_V X P \text{ since } P_V^T P = P_V^T} \text{Softmax}[U[XP]^T Q[XP]] \\ &= (P_V^T + \Omega_V X) P \end{aligned}$$

$$= U[X] P \text{Softmax}[P^T U[X]^T Q[X] P]$$

$$\begin{aligned} &= U[X] \underbrace{P P^T}_{=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}} \text{Softmax}[P^T U[X]^T Q[X] P] P \\ &= \text{Softmax}[U[X]^T Q[X]] P = S_a[X] P \end{aligned}$$

$$12.4 \quad y_i = \text{softmax}_i[\mathbf{z}] = \frac{\exp[\mathbf{z}_i]}{\sum_{j=1}^5 \exp[\mathbf{z}_j]}$$

$$\frac{\partial y_i}{\partial z_j} = \begin{cases} \text{softmax}_i[\mathbf{z}] (1 - \text{softmax}_i[\mathbf{z}]) & i=j \\ -\text{softmax}_i[\mathbf{z}] \cdot \text{softmax}_j[\mathbf{z}] & i \neq j \end{cases}$$

$$\text{so: } \frac{\partial y_1}{\partial z_1} = \frac{\exp[-3]}{\exp[100]} \left(1 - \frac{\exp[-3]}{\exp[100]} \right) \text{ as } \sum_{j=1}^5 \exp[\mathbf{z}_j] \approx \exp[\mathbf{z}_3] = \exp[100]$$

$$\frac{\partial y_1}{\partial z_2} = \frac{\exp[1]}{\exp[100]} \cdot \frac{\exp[-3]}{\exp[100]}$$

and so on

12.5 We can then compute all heads at once with one matrix multiply!

12.6 predict masked words: generative, probably for syntax and 'understanding'

classifying pairs of sentence as being adjacent: contrastive, probably for learning 'flow' of text / structure

e.g., two sentences: one original, one permuted.

classify if permutation changed meaning (synonyms vs. random changes)

classifying sentence as being part of the same document

12.7 Essentially one extra row of attention is computed!

12.8 Similarly to convolution operations, we could use a sparse interaction matrix.

Alternatively, we could decrease the size of the attention matrix.

12.9 Channel-wise attention should require twice the computation of patchwise attention:

$2 \cdot (16 \cdot 16)^2 \cdot 512$ between patches
vs $2 \cdot 512^2 \cdot 256$ between channels

$$12.10 \quad a[x_m, x_n] = \text{softmax}_n [e_m^T q_n] = \frac{\exp[e_m^T q_n]}{\sum_{m'=1}^N \exp[e_{m'}^T q_n]}$$

If we replace $\exp[e_{m'}^T q_n]$ with $g[e_{m'}]^T g[q_n]$:

$$\sum_{m'=1}^N (g[e_{m'}]^T g[q_n]) = \left(\sum_{m'=1}^N g[e_{m'}]^T \right) g[q_n]$$

Thus we go from $O(ND)$ to $O(D')$